Conservation of Energy – Systems with Several Objects

Answer key

Conventions used throughout

• **Energy symbols (from the lessons):** W = work, E_k = kinetic energy ($\frac{1}{2}mv^2$), E_g = gravitational potential energy (mgh), E_e = elastic potential energy ($\frac{1}{2}kx^2$). x is the stretch or compression measured from the spring's unstretched length.

• **Method (from the lesson):** draw the situation, list the kinds of energy at the start and at the end, then write $E_{TOTi} = E_{TOTf}$. With friction or another non-conservative force: $E_{TOTi} - W_F = E_{TOTf}$, where W_F is the size of the work done by that force.

• **Heights:** E_g is relative, so each problem picks its own h = 0 (usually the lowest point of the object).

• **Several objects:** the total energy of the whole system (hanging mass + cart + spring) is conserved. Each object gets its own h = 0 at its lowest point, as the lesson says. The string does not stretch, so both objects move the same distance and have the same speed.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• **Final answers:** work and energy are scalars (a size and a sign, no direction). Vectors are given as a size with the direction in square brackets.

• g = 9.81 m/s^2. Every step keeps full precision; only final answers are rounded.

EXAMPLE 1, QUESTION 1

How each energy changes

Consider the following system. It is initially held in place and the spring is not stretched or compressed. After the cart is released, it rolls up the ramp and the hanging mass falls down a distance h before the system comes to rest again. The spring has no mass.

1. When comparing the **initial** and **final** state of the system, record how each type of energy has changed by writing "gain" "loss" or "no change" in the table.

The system starts at rest and ends at rest, so nothing has kinetic energy at the start or at the end. The hanging mass ends lower, the cart ends higher, and the spring ends stretched.

	Gravitational Potential	Elastic Potential	Kinetic Energy
Hanging Mass	loss (it ends lower)	N/A	no change (at rest before and after)
Cart	gain (it ends higher)	N/A	no change (at rest before and after)
Spring	no change (it has no mass)	gain (it ends stretched)	no change (it has no mass)

FINAL ANSWER

Hanging mass: E_g loss, E_k no change

Cart: E_g gain, E_k no change

Spring: E_g no change, E_e gain, E_k no change

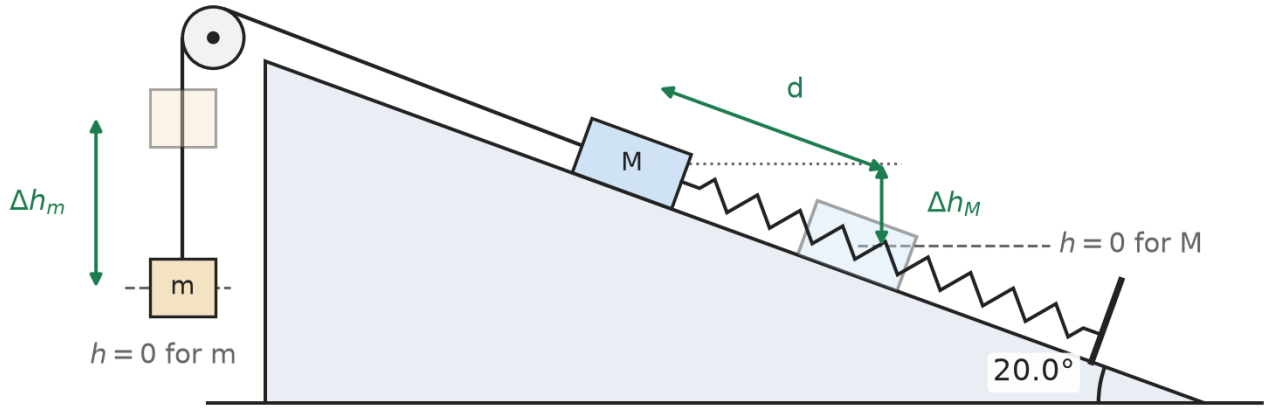
EXAMPLE 1, QUESTION 2

Labelling h = 0 and the heights

2. Because gravitational potential energy is relative, we are only looking at gains or losses for a particular object. Because of this, we can choose h=0 to be different for the two objects; we will set h=0 to be the lowest point for each object. Hence, h=0 for the hanging mass will be its ending point and h=0 for the cart will be its starting point. **Label** these on the diagram as well as Δh_m (for the hanging mass) and Δh_M (for the cart).

Each object's h = 0 is at its lowest point, and each Δh is measured straight up or down from there.

The system with h = 0, Δh_m and Δh_M labelled



faded = starting positions (spring unstretched); solid = final positions

The hanging mass m falls Δh_m to its h = 0. The cart M starts at its h = 0, moves a distance d up the ramp and rises Δh_M.

FINAL ANSWER

h = 0 for m: its final (lowest) position. Δh_m is the distance it falls.

h = 0 for M: its starting (lowest) position. Δh_M is the vertical height it gains.

EXAMPLE 1, QUESTION 3

Comparing the distances

3. How does the distance the mass falls compare with the distance the cart moves along the ramp and the amount the spring stretches?

They are all the same. The string does not stretch, so when the mass falls a distance Δh_m the cart is pulled the same distance d along the ramp. The spring is attached to the cart and started unstretched, so its stretch x is that same distance.

FINAL ANSWER

Δh_m = d = x (all three are equal)

EXAMPLE 1, QUESTION 4

Comparing the heights

4. How does the height the hanging mass falls compare with the height the cart gains? Write an equation to relate them.

The cart moves a distance d = Δh_m along the ramp, but only part of that is vertical. In the right triangle made by the ramp, d is the hypotenuse and Δh_M is the side opposite the ramp angle θ.

Sine of the ramp angle: $\sin\theta = \dfrac{\Delta h_M}{d}$

Multiply by d, and put in $d = \Delta h_m$: $\Delta h_M = \Delta h_m \sin\theta$

The cart gains less height than the hanging mass loses, because sin θ is less than 1.

FINAL ANSWER

$\Delta h_M = \Delta h_m \sin\theta$ (the cart gains less height than the mass falls)

EXAMPLE 1, QUESTION 5

How far the hanging mass falls

5. Determine how far the hanging mass fell if the mass of the cart is 500. g, the hanging mass is 300. g, the spring has a force constant of 30.0 N/m and the ramp makes a 20.0° angle to the horizontal. Friction is negligible.

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$M = 500.\ \text{g} = 0.500\ \text{kg}$ $\quad$ $m = 300.\ \text{g} = 0.300\ \text{kg}$

$k = 30.0\ \text{N/m}$ $\quad$ $\theta = 20.0°$

$h = ?$

Call the distance the hanging mass falls h. From questions 3 and 4, the spring stretches x = h and the cart rises h sin θ. The system is at rest at the start and at the end, so there is no kinetic energy in either state.

Step 1 – Energy equation

Start: only the hanging mass has energy (it is h above its h = 0). End: the cart has gravitational energy and the spring has elastic energy.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $mgh = Mg(h\sin\theta) + \frac{1}{2}kh^2$

Step 2 – Isolate h

Every term has an h. h = 0 is the starting position, so divide it out to get the other answer.

Divide both sides by h: $mg = Mg\sin\theta + \frac{1}{2}kh$

Subtract $Mg\sin\theta$, multiply by $\frac{2}{k}$: $h = \frac{2g\,(m - M\sin\theta)}{k}$

Put in the numbers: $h = \frac{2(9.81)(0.300 - 0.500\sin 20.0°)}{30.0}$

$$h = \frac{2(9.81)(0.300 - 0.17101)}{30.0} = 0.08436 \text{ m}$$

FINAL ANSWER

h = 0.0844 m = 8.44 cm

Worksheet answer: 8.4 cm using g=9.81 N/m or 8.6 using g=10 N/m
Why the answers differ: Same value, different rounding. The key uses g = 9.81 and gets 8.436 cm. The data (500. g, 300. g, 30.0 N/m, 20.0°) have 3 significant figures, so the key gives 8.44 cm; the worksheet rounds to 2.

EXAMPLE 1, QUESTION 6

Speed after falling 4.0 cm

> 6. How fast is the hanging mass moving when it has fallen 4.0 cm? (hint: which objects have kinetic energy? How do the two speeds compare?)

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$d = 4.0\text{ cm} = 0.040\text{ m}$	$M = 0.500\text{ kg}$
$m = 0.300\text{ kg}$	$k = 30.0\text{ N/m}$
$\theta = 20.0°$	$v = ?$

Both the hanging mass and the cart are moving, so both have kinetic energy. They are joined by the string, so they have the same speed v. This time take h = 0 for the hanging mass at the 4.0 cm point, so it starts d = 0.040 m above its h = 0.

Step 1 – Energy equation

Start: the hanging mass is d above its h = 0. After 4.0 cm: the cart has risen d sin θ, the spring is stretched d, and both objects move at speed v.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $$mgd = Mg(d\sin\theta) + \frac{1}{2}kd^2 + \frac{1}{2}mv^2 + \frac{1}{2}Mv^2$$

Step 2 – Isolate v

Collect the kinetic energy:

$$\frac{1}{2}(m+M)v^2 = mgd - Mgd\sin\theta - \frac{1}{2}kd^2$$

Multiply by $\frac{2}{m+M}$, square root:

$$v = \sqrt{\frac{2\left(mgd - Mgd\sin\theta - \frac{1}{2}kd^2\right)}{m+M}}$$

Step 3 – Put in the numbers

Hanging mass: $mgd = (0.300)(9.81)(0.040) = 0.11772$ J

Cart: $Mgd\sin\theta = (0.500)(9.81)(0.040)\sin 20.0° = 0.06710$ J

Spring: $\frac{1}{2}kd^2 = \frac{1}{2}(30.0)(0.040)^2 = 0.02400$ J

Speed: $v = \sqrt{\frac{2(0.11772 - 0.06710 - 0.02400)}{0.300 + 0.500}} = 0.2580$ m/s

FINAL ANSWER

v = 0.26 m/s (the cart moves at the same speed)